WORKED SOLUTION

STEP III Specimen paper, Question 9

Pure • Groups of Matrices

STEP III Specimen paper, Question 9

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/spec-s3-q9

The key idea

Everything follows from one calculation: multiply two general matrices of this shape and see what comes out. The product turns out to have the same shape, with entries given by a simple formula, and that formula hands us closure, the identity, the inverse, the integer subgroup and the commuting condition almost for free. Write

$$M(x,y,z) = \begin{pmatrix} 1 & 0 & 0 \\ x & 1 & 0 \\ y & z & 1 \end{pmatrix}.$$

Solution

G is a group

The product formula. Multiplying row by column,

$$M(a,b,c)\,M(x,y,z) = \begin{pmatrix} 1 & 0 & 0 \\ a & 1 & 0 \\ b & c & 1 \end{pmatrix}\begin{pmatrix} 1 & 0 & 0 \\ x & 1 & 0 \\ y & z & 1 \end{pmatrix} = \begin{pmatrix} 1 & 0 & 0 \\ a+x & 1 & 0 \\ b+cx+y & c+z & 1 \end{pmatrix},$$

that is,

$$M(a,b,c)\,M(x,y,z) = M(a+x,\ b+cx+y,\ c+z). \qquad (*)$$

Closure. The entries $a+x$, $b+cx+y$ and $c+z$ are real, so the product lies in G.

Associativity. We are allowed to assume this for matrix multiplication.

Identity. The identity matrix is $I = M(0,0,0)$, which is in G, and $IM = MI = M$ for every matrix M.

Inverses. Given $M(a,b,c)$, we look for $M(x,y,z)$ with product $M(0,0,0)$. By $(*)$ we need $a+x=0$, $c+z=0$ and $b+cx+y=0$, so $x=-a$, $z=-c$ and $y=-b-cx=ac-b$. Check the other order too, because matrix multiplication isn't commutative:

$$M(-a,\ ac-b,\ -c)\,M(a,b,c) = M(0,\ (ac-b)+(-c)a+b,\ 0) = M(0,0,0).$$

So $M(a,b,c)^{-1} = M(-a,\ ac-b,\ -c)$, which is in G.

All four group axioms hold, so G is a group under matrix multiplication.

The integer matrices

Let H be the set of $M(x,y,z)$ with $x,y,z \in \mathbb{Z}$. It is a subset of G and contains $I = M(0,0,0)$. If a,b,c,x,y,z are integers then so are $a+x$, $b+cx+y$ and $c+z$, so by $(*)$ H is closed under multiplication. The inverse $M(-a,\ ac-b,\ -c)$ also has integer entries, so H is closed under inverses. Associativity is inherited from G. So **yes**, H is a subgroup of G.

? **Ask yourself** Why isn't closure under multiplication enough on its own?

For an infinite subset it can fail: the matrices $M(x,y,z)$ with x,y,z *non-negative* integers are closed under multiplication, but $M(1,0,0)$ has inverse $M(-1,0,0)$, which is not in that set. Here the inverse formula keeps us inside the integers, and that is what makes H a subgroup.

An element commuting with everything

Take $A = M(a,b,c)$ and a general $B = M(x,y,z)$. By $(*)$, in both orders,

$$AB = M(a+x,\ b+cx+y,\ c+z), \qquad BA = M(x+a,\ y+za+b,\ z+c).$$

The first and third entries always agree, so

$$AB = BA \iff cx = az.$$

(In particular G is not abelian: $M(1,0,0)$ and $M(0,0,1)$ don't commute.)

For A to commute with *every* B we need $cx = az$ for all real x and z. Choosing $x=1$, $z=0$ gives $c=0$; choosing $x=0$, $z=1$ gives $a=0$. Conversely, if $a=c=0$ then $cx = 0 = az$ for all x,z. So the elements that commute with all of G are exactly

$$A = M(0,b,0) = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ b & 0 & 1 \end{pmatrix}, \qquad b \in \mathbb{R}.$$

Yes, such an A exists: any $b \neq 0$ gives $A \neq I$, for example $A = M(0,1,0)$. By $(*)$, $M(0,1,0)\,M(x,y,z) = M(x,\ 1+y,\ z) = M(x,y,z)\,M(0,1,0)$ for every $B = M(x,y,z)$ in G.

Checks

- Each matrix in G is lower triangular with ones on the diagonal, so it has determinant 1 and is certainly invertible: our inverse must exist, and it is again of the same shape.
- Numerically: with $A = M(2,3,5)$ and $B = M(1,0,0)$, $(*)$ gives $AB = M(3,8,5)$ but $BA = M(3,3,5)$, so A does not commute with B, as $cx = 5 \neq 0 = az$ predicts.

We also checked the product and inverse formulas, and the commuting condition, exactly with fractions for thousands of random matrices.

Common slips

- Checking the inverse on one side only. In a non-abelian setting it is worth confirming both products (or noting that for square matrices a one-sided inverse is automatically two-sided).
- Answering the final part with "G is not abelian, so no". A non-abelian group can still have non-identity elements commuting with everything, and this one does.
- Forgetting the converse: after showing $a = c = 0$ is necessary, we must also check that such A really does commute with every B.

Beyond STEP

This is the *Heisenberg group*, and the elements commuting with everything form its *centre* $Z(G)$, here the matrices $M(0, b, 0)$, a copy of the real numbers under addition. The integer version H is the discrete Heisenberg group, a favourite example in group theory and geometry, and the Heisenberg group itself appears in quantum mechanics through the commutation relations of position and momentum. For groups, subgroups and centres, see Allenby, *Rings, Fields and Groups*.

Beyond STEP worked solutions: https://beyondstep.keithmcnulty.org/. Solution text CC0 1.0; the question itself is © Cambridge University Press & Assessment and is not reproduced here.